



Pole Fabrication Checklist

Incorporating:

BT
CityFibre
REC
NBI
ANSI (SYP)

Ref no:	QTG116	Calders & Grandidge	
Compiled By: D. Losty	Quality Training Guide Pole Fabrication Checklist		Revision 13 Revision Date: Oct 2025
Effective Date: Oct 2019			Revised By: D. Losty

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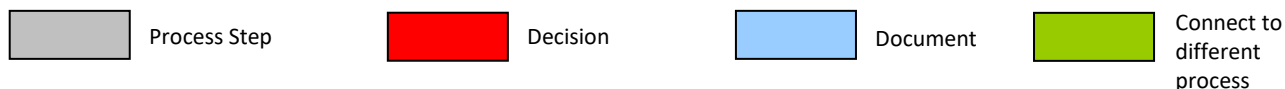
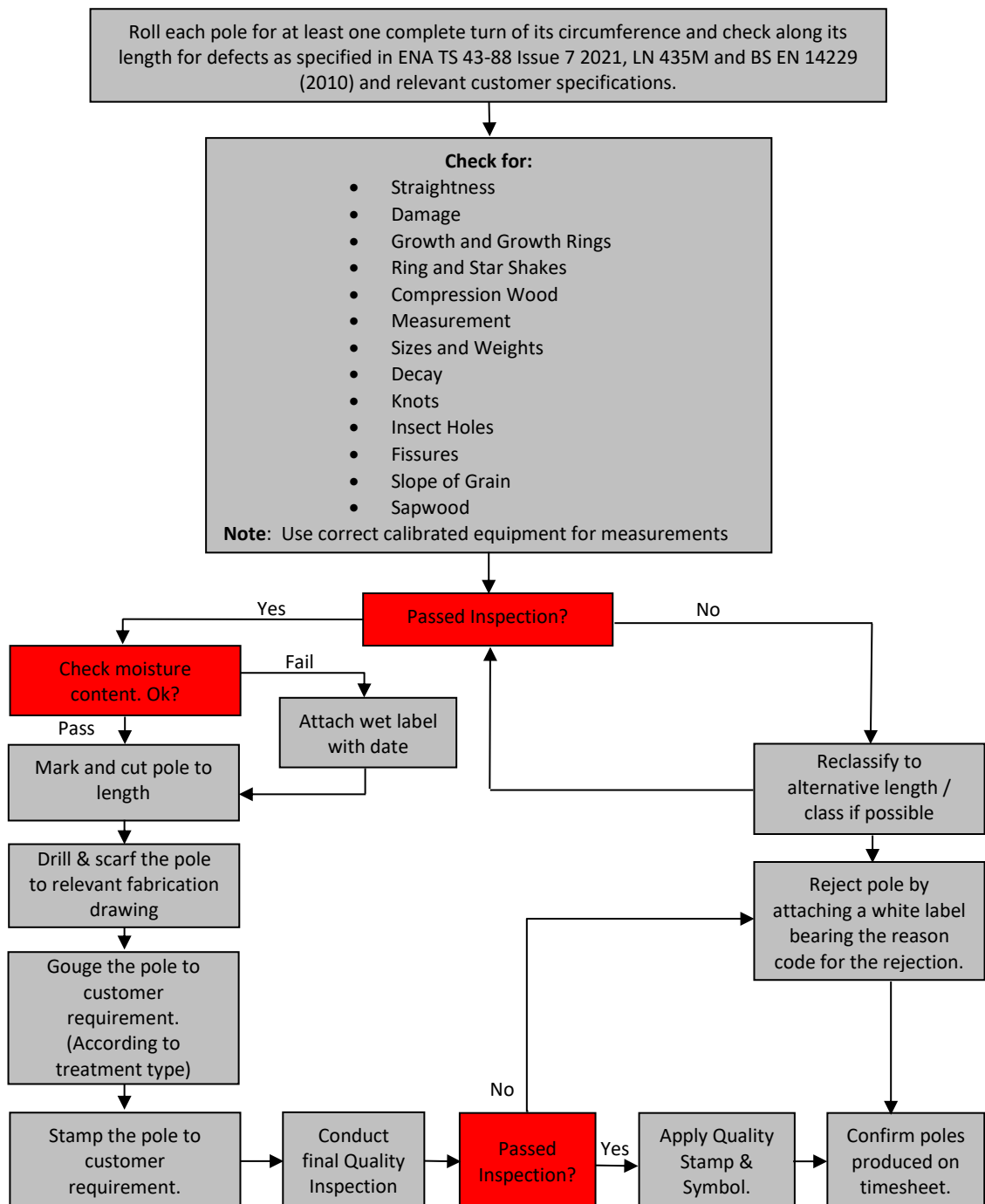
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PROCESS FLOW

This document details the inspection and fabrication process for wood poles. Information taken from Customer specifications, ENA TS 43-88 Issue 7 2021, LN 435M and BS EN 14229 (2010).

NO DEVIATION FROM THE SPECIFIED REQUIREMENTS IS PERMITTED WITHOUT PRIOR WRITTEN AGREEMENT FROM THE CUSTOMER.



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Key	
BASE	Minimum pole requirements from BS EN 14229:2010 & BS1990-1:1984.
REC	Additions/exceptions to BASE for Regional Electric Companies.
BT	Additions/exceptions to BASE for British Telecom.
NGED	Additions/exceptions to BASE for National Grid Energy Distribution (previously WPD).
CFR	Additions/exceptions to BASE for CityFibre.
ENW	Additions/exceptions to BASE for Electricity North West.
NBI	Additions/exceptions to BASE for National Broadband Ireland.
SYP	Specific ANSI requirements for Southern Pine poles.

1 TREATMENT RESTRICTIONS

- 1.1 **ENW** Preservation treatment using water-borne salts must not be used. **All** Electricity North West's poles must be treated with creosote.
- 1.2 **BT** Unless specially advised by BT, no AC500 poles should be supplied to BT.
- 1.3 **REC** Both creosote treated poles and treatment using water-borne salts are permissible.
- 1.4 **NGED** Treatment using AC500 is not permissible.
- 1.5 **NBI** Both creosote treated poles and treatment using water-borne salts are permissible.
- 1.6 **SYP** Must be treated to the specific parameters laid out in QF112.

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2 STRAIGHTNESS

2.1

BASE

A single sweep is permitted to the extent that the longitudinal axis of the wood pole remains within a distance of 1% of the pole length from a straight line drawn from the centre of the tip to the centre of the pole 1.5m from the butt. See Diagram 1 below.

A double sweep or short crook is permitted to the extent that a straight line from the centre of the top to the centre of the wood pole 1.5 m from the butt remains within the wood pole. See Diagram 2 below.

2.2

BT
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CFR

The pole should have a uniform taper. Short crooks, double sweeps, butt sweeps are not allowed. Butt flutes are allowed within the limitations on size (see Size and Weights). A single sweep is allowed to the extent that a straight line from the centre of the butt to the centre of the tip does not at any intermediate point pass nearer to the surface of the pole than one quarter of the diameter at the point in question. See Diagram 1 below.

A reverse sweep where the sweep is in 2 directions and one plane is allowed, to the extent that a straight line from the centre of the butt to the centre of the tip does not at any intermediate point pass nearer to the surface of the pole than $\frac{1}{3}$ of the diameter at the point in question. See Diagram 2 below.

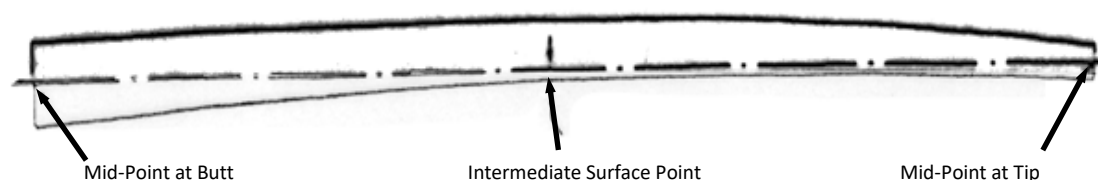


Diagram 1 – Measurement of sweep in one plane and one direction

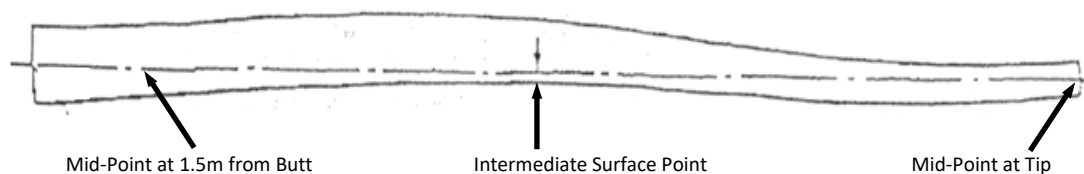


Diagram 2 – Measurement of reverse sweep in two directions and one plane

2.5

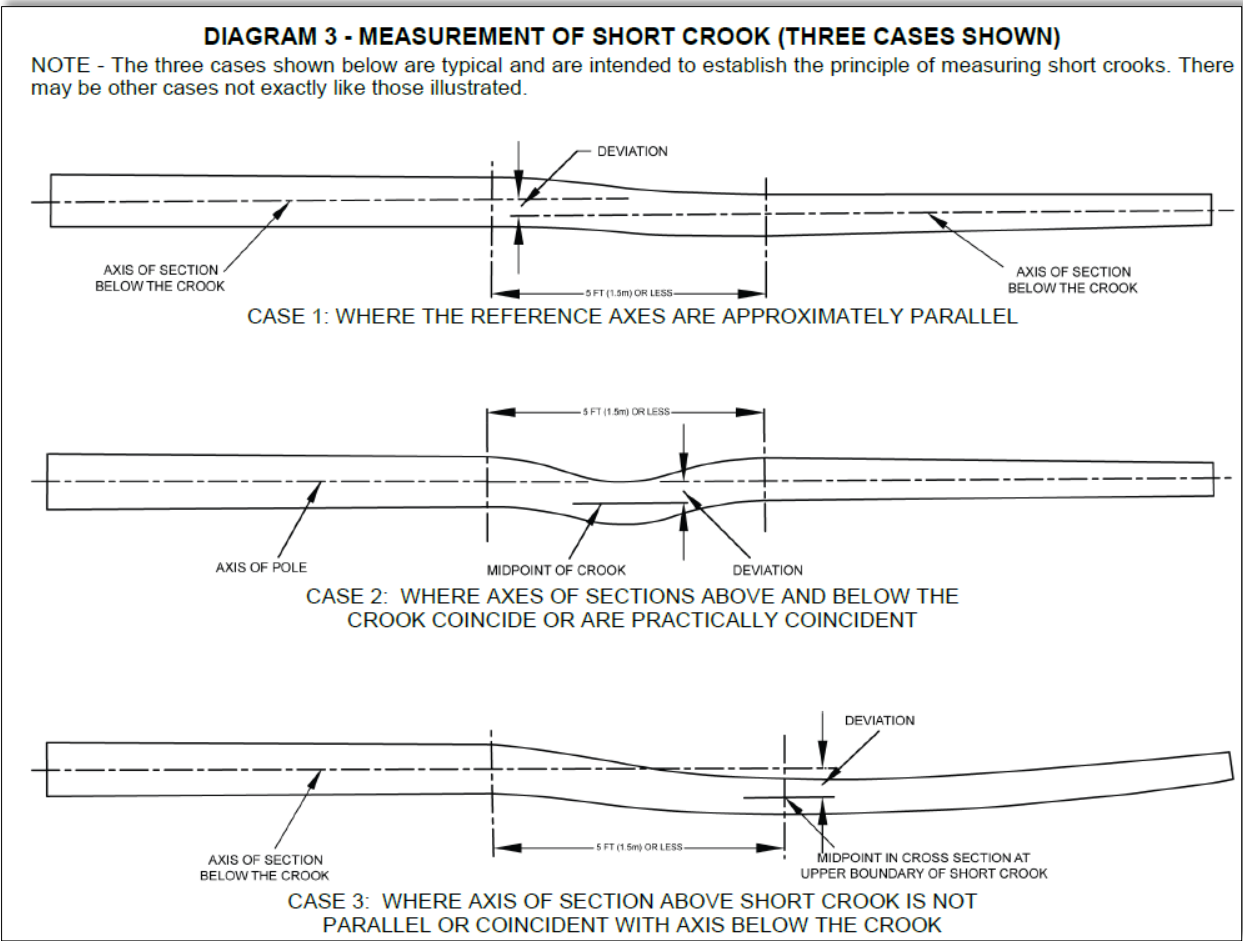
NGED

A double sweep or a short crook is not permitted.

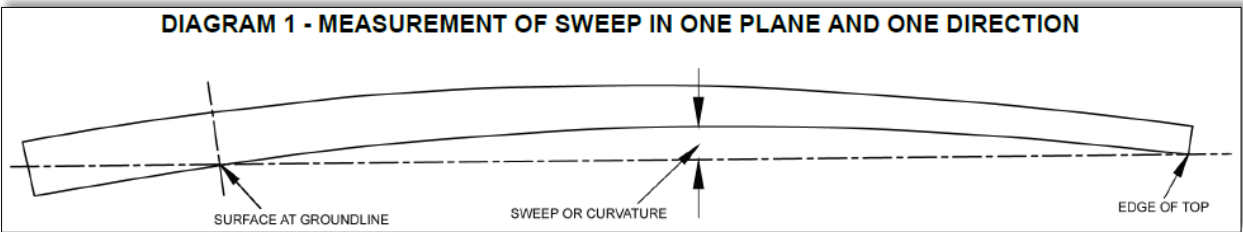
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Short crooks are not permitted. A short crook is defined as follows:

- 2.6 **SYP** A localized deviation from straightness that, within any section that is 1.5m or less in length, is more than 1/2 the mean diameter of the crooked section. (See Diagram 3 below).



- 2.7 **SYP** Where sweep is in one plane and one direction only - a straight line joining the surface of the pole at the groundline and the edge of the pole at the top must not be distant from the surface of the pole at any point by more than 25mm for each 3m in length. (See Diagram 1 below.)

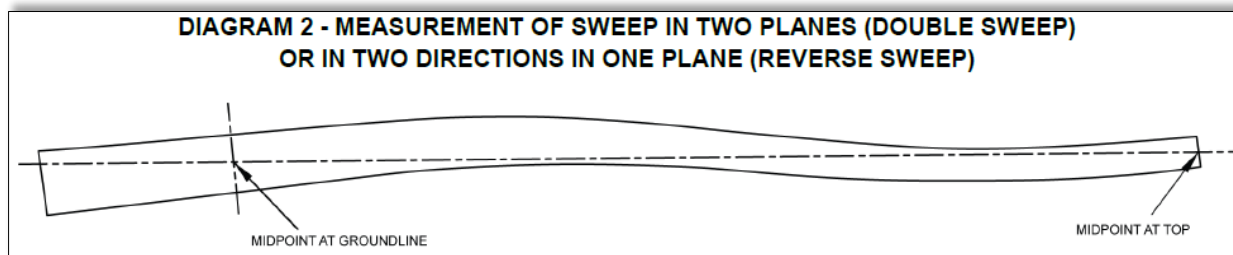


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2.8

SYP

Where sweep is in two planes (double sweep) or in two directions in one plane (reverse sweep) - a straight line connecting the midpoint at the groundline with the midpoint at the top must not at any intermediate point pass through the surface of the pole. (See Diagram 2 below.)



3. DAMAGE

3.1

BASE

The wood pole diameter on which the measurement of the damage is based must be calculated on the nominal diameter at the cross section where the damage occurs. To determine the nominal diameter, the nominal diameter of the sound wood pole immediately above and below the damage must be measured and averaged. The minimum diameter of the damaged cross section must be measured and the reduction in diameter determined.

Mechanical damage must not extend to a depth of the wood pole that reduces the diameter by more than 5% of the diameter at any cross section.

No more than two occurrences of mechanical damage are permitted and no part of these can be less than 500 mm apart.

3.2

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Mechanical damage is limited to broken surface fibres whose depth must not exceed 5mm and where the damage must be removed by trimming.

Rind galls, animal bites, bird damage and inbarks with a depth of 5mm or less may be removed.

Removal the above defects is restricted to 2 occurrences on a pole, and they must **not** be on the same cross section (causing waisting of the pole).

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3.3

SYP

Poles are not acceptable if they contain indentations attributed to loading or handling slings that are $\geq 6\text{mm}$ deep over 20% or more of the pole circumference, or $>13\text{mm}$ deep at any point.

Other indentations or abrasions, for example, forklift damage, kiln sticker damage, chain-saw damage, etc., must not be more than $1/10$ the pole diameter at the point of damage up to a maximum of 25mm. Such damage is permitted in an oversized section, where the excess of wood must be taken into consideration in evaluating the effects of the damage.

In any case, the circumference for a given class is still required to be not less than the specification minimum.

4 BARK POCKETS & RINDGALLS

4.1

BASE

Bark pockets and rind galls are permitted without limitation in the first 1.0m of length from the butt.

Above the first 1.0m length from the butt, bark pockets and rind galls must be measured as the overall length and width at the widest point and depth at the deepest point.

Rind galls above the first 1.0m from the butt are allowed with the following restrictions on size:

Length ≤ 0.4 times the nominal pole diameter.
Width ≤ 0.5 times the nominal pole diameter.
Depth ≤ 0.05 times the nominal pole diameter.

For example: A pole with a nominal diameter of 300mm at the defect would be allowed a maximum size of rind gall of 120mm long, 150mm wide and 15mm deep.

Bark pockets and rind galls must not extend to a depth of the wood pole that reduces the diameter by more than 5% of the diameter at any cross section.

4.2

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Rind galls within 900mm of the butts are allowed with the following restrictions on size:

Length $\leq 1 \times$ Diameter of pole at mid-point of defect
Width $\leq 0.25 \times$ Diameter of pole at mid-point of defect
Depth $\leq 0.25 \times$ Diameter of pole at mid-point of defect

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For example: A pole with a diameter of 300mm at the mid-point of the defect would be allowed a maximum size of rind gall of 300mm long, 75mm wide and 75mm deep.

Depressions in excess of 5mm in depth are not permitted.

Compressed wood caused by open stacking must be limited to no more than a total decrease in diameter of 5mm measured across the pole.

4.3

NGED

Bark pockets and rind galls shall be subject to the following dimensional limitations along the entire length of the pole:

- a) Length — 0.4 times the nominal pole diameter.
- b) Width — 0.5 times the nominal pole diameter.
- c) Depth — 0.05 times the nominal pole diameter.

4.4

SYP

Depressions containing bark inclusions must not be more than 50mm in depth, measured from the surface of the pole.

5 COMPRESSION WOOD

5.1

BASE

Compression Wood is not permitted.

5.2

BASE

A movement of the pith in excess of 25 mm from centre is considered a failure.

5.3

SYP

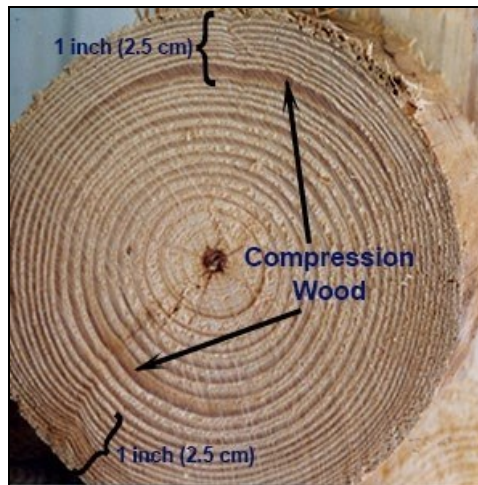
The outer 25mm of all poles shall be free from visible compression wood.

Compression wood is defined as abnormal wood formed on the lower side of branches and inclined trunks of softwood trees. Compression wood is identified by its relatively wide annual rings, usually eccentric; relatively large amount of latewood (summerwood), sometimes more than 50% of the width of the annual rings in which it occurs; and its lack of demarcation between earlywood (springwood) and latewood (summerwood) in the same annual rings. Compression wood, compared with normal wood, shrinks excessively lengthwise.

In softwoods, compression wood is formed on the lower side of a branch or on the lower side of the main stem of a leaning tree. Because the fibril angle of compression wood is at a steeper angle than that of normal wood,

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compression wood shrinks at a greater rate in the longitudinal dimension than does normal wood, usually causing the surface of the pole to loosen or separate from the rest of the pole. For this reason, compression wood is not allowed within 2.5cm of the surface when measured at either end of the pole. This photograph shows compression wood as it sometimes occurs in southern pine. This pole must be rejected because the dark, slightly eccentric ring in the upper half of the pole top is within 2.5cm of the surface of the pole.



The following photograph shows how compression wood can cause cross breaks. Note the buckling that occurs on the surface due to the shrinkage of the underlying compression wood. This pole must be rejected.



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6 SAPWOOD

- 6.1 **BASE** The minimum depth of sapwood for all pole sizes is 25mm (around the full circumference of the pole, after dressing).

7 RATE OF GROWTH

- 7.1 **BASE**
- Rate of growth must be measured at either the tip or butt of the wood pole and expressed as the mean number of growth rings per 25mm.
- Sometimes, there is little colour difference between early and late growth. As a rule, more growth occurs during the summer (known as early wood) and less growth during the winter (known as late growth). The wide spaces are early wood and the narrow spaces are late wood. Therefore, one growth ring is a combination of one ring of early wood and one ring of late wood combined.
- The measurements must be made over a radial line, as long as possible, commencing 50mm from the pith.
- For wood poles that have a theoretical diameter of less than 150mm, measurement must be made over a radial line as long as possible commencing from the circumference.
- The growth rings should be closely pitched with Pinus Sylvestris having a minimum of 8 rings per 25mm.

- 7.2 **BT & CFR**
- At the tip of the pole the distance from the pith to the 12th growth ring must not exceed ½ of the pole diameter.
- The growth rings should be closely pitched with Pinus Sylvestris having a minimum of 8 rings per 25mm.

- 7.3 **NGED**
- Measurements must be taken at the tip within 50mm of the circumference.
- The diameter of the juvenile wood must not exceed 50% of the pole diameter when measured at the tip. Juvenile wood is the area inside the 12th ring from the pith.

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7.4

SYP

The average rate of growth must be not less than 6 rings per 25mm when measured:

(a) on the butt in the outer 50mm of poles having a minimum required circumference of 950mm or less at 1.8m from the butt, and

(b) in the outer 75mm of poles having a minimum required circumference of more than 950mm at 6 feet (1.8m) from the butt.

For poles that exhibit a non-uniform growth rate around the circumference, the average growth must be determined at the midpoint of the shortest arc between the point showing the fewest growth rings in the required zone and the point showing the most growth rings in the required zone.

Exception: Poles with an average of as few as 4 rings per 25mm in the required zone are acceptable if 50% or more latewood (summerwood) is present in the zone where the average rate of growth is determined.

As an alternative, or in cases where the determination in the butt is uncertain, the ring count and latewood (summerwood) measurements mentioned above may be made on an increment core taken at 1.8m from the butt directly above the place where the average rate of growth is indicated on the butt surface.

8 SHAKES

8.1

BASE

The tip must be free from ring or star shakes with three or more points.

At the butt, one complete ring or star shake is acceptable, provided that not more than two points extend to within 5mm of the pole circumference. If they extend to the circumference, they must not extend along the pole more than 500mm from the butt.

No more than one point must extend to the surface.

8.2

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The tip must be free from ring or star shakes but one complete ring or one star shake is acceptable at the butt provided that no part extends to within 50mm of the periphery.

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Shakes in the butt surface that are not closer than 50mm to the side surface of the pole are permitted, provided they do not extend to the groundline.

Shakes or a combination of connected shakes that are closer than 50mm to the side surface of the pole are permitted, provided they do not extend farther than 610mm from the butt surface and do not have an opening wider than 3mm.

Shakes in the top surface are permitted in poles that are to be given full-length preservative treatment, provided that the shake is not closer to the surface of the pole than the midpoint of a line extending from the pith to the surface (i.e., the shake is permitted if it is closer to the pith than to the surface of the pole).

Explanation:

A shake is a longitudinal separation between the annual rings of a tree, as is shown in this southern pine pole.

- 8.3 **SYP** Since this shake is closer than 5cm to the surface of the pole, it must have an opening less than 3mm wide to be accepted. If it is less than 3mm wide, it cannot extend farther than 610mm from the butt of the pole.



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9 MEASUREMENT

9.1

BASE

All measurements must be made when the wood pole is at or above fibre saturation point.

The taper of wood poles covered by BS EN 14229 is expected to be between 6 mm and 16 mm per metre length.

Length

Length of the wood pole must be measured using a tape measure. Where one or both ends of the wood pole are not cut square, the minimum length must be recorded.

Tolerance: -1% / +2%

Diameter

Maximum and minimum diameters must be measured using callipers. Alternatively, the theoretical diameter may be calculated from the circumference measured by using a tape measure.

Tolerance: -0mm / +40mm, unless otherwise declared by the manufacturer.

NOTE: A list of commonly used wood pole sizes (minimum nominal diameter at 1.5m from the butt and length) is given in APPENDIX 3.

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9.2

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Minimum Diameters

Homegrown – The diameter must be measured by callipering in 2 directions at right angles to each other, the **minimum** diameter obtained is valid.

Imported – The diameter must be measured by callipering in 2 directions at right angles to each other, the **average** diameter obtained is valid.

Maximum Diameters.

The diameter must be measured by callipering and the **maximum** diameter of the pole at that cross section is valid.

Sizes and Weights

After felling, conversion and final dressing (i.e. machine dressing at Pole Suppliers Depot), the pole must be in accordance with the dimensions for its class as shown in Tables BT1, BT2 and BT3.

The diameter at top must be measured at the bolt hole as detailed in Drawing CN206. The tolerance on length is -50/+150mm.

Butt flutes must not cause the minimum diameter at the butt to be less than that required at 1.5m. Maximum butt sizes must be such that the butt of the pole will fit into a circular hole of diameter 300mm for light poles, 350mm for medium poles and 400mm for stout poles.

On delivery poles must not exceed the weights shown in Tables BT1, BT2 & BT3.

TABLE BT1

SIZES AND WEIGHTS OF LIGHT POLES						
LENGTH	ITEM CODE No's	DIAMETER AT TOP		MINIMUM DIAMETER AT 1.5M FROM BUTT END	MAXIMUM WEIGHT IMP. RED.	NOMINAL VOLUME
		MIN	MAX			
		mm	mm	mm	kg	m ³
6	013041	125	175	150	112	0.110
7	013042	125	175	160	134	0.129
8	013043	125	175	170	165	0.199
9	013044	125	175	180	203	0.225
10	013045	125	175	185	245	0.270
11	013046	125	175	195	276	0.320
13	013047	130	175	210	391	0.430

BT
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CFR

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TABLE BT2

SIZES AND WEIGHTS OF MEDIUM POLES						
LENGTH	ITEM CODE No's	DIAMETER AT TOP		MINIMUM DIAMETER AT 1.5M FROM BUTT END	MAXIMUM WEIGHT IMP. RED.	NOMINAL VOLUME
		MIN	MAX			
		mm	mm	mm	kg	m ³
7	013048	140	175	200	194	0.210
8	013049	145	200*	210	230	0.240
9	013050	150	200*	220	275	0.315
10	016439	150	200	230	329	0.360
11	013051	150	200	240	383	0.430
12	013052	150	200	250	457	0.490
13	013053	160	200	260	625	0.576
15	013054	165	210	290	697	0.765

* We have dispensation from BT on these sizes. Normally 180mm.

TABLE BT3

SIZES AND WEIGHTS OF STOUT POLES						
LENGTH	ITEM CODE No's	DIAMETER AT TOP		MINIMUM DIAMETER AT 1.5M FROM BUTT END	MAXIMUM WEIGHT IMP. RED.	NOMINAL VOLUME
		MIN	MAX			
		mm	mm	mm	kg	m ³
9	013048	190	250	275	379	0.460
10	013049	190	250	285	430	0.550
11	013050	190	250	295	510	0.653

Imported European Redwood and UK Grown Scots Pine

9.3

ENW

Light Duty Wood Poles are not required by Electricity North West.

Scots Pine (Pinus Sylvestris) must be used.

In addition to the sizes given in ENA TS 43-88, Appendix A:

The sizes classed as 'Extra Stout', and listed in Table 1 below, must be available.

Sizes classed as "Extra-Extra Stout" are also available, but these are rarely used. Therefore, if required, they will be ordered on a bespoke basis – specification to be agreed in each case.

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Nominal length (m)	Diameter at top		Minimum diameter at 1.5m from butt end (mm)
	Min(mm)	Max(mm)	
9	190	250	295
10	195	250	305
11	200	250	315
12	200	255	325
13	200	265	340
14	200	265	355
15	200	265	370
16	205	270	385
17	210	275	395
18	210	275	410
20	215	275	435
22	215	275	455
24	215	275	490
NOTE: Maximum top diameters are given only to indicate the range of taper to be expected.			

9.4

NBI

Poles supplied to NBI are required to have a measurement at the extreme butt of the pole of $\leq 330\text{mm}$ (in order to fit into holes drilled by a 350mm auger).

9.5

NGED

Poles supplied to NGED are subject to *nominal* maximum dimensions, as follows:

- The nominal maximum dimension (for girth and/ or tip size) is the stated minimum dimension + 40mm.

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Length

Poles less than 15.2m in length must be not be more than 150mm longer than nominal length.

Poles 15.2m or more in length must be not more 310mm longer than nominal length.

Length must be measured between the extreme ends of the pole.

Diameter

General

Poles are classed while in the green condition, after bark removal and/or shaving. Subsequently, there may be some shrinkage due to conditioning, seasoning, or while in service. Therefore, this shrinkage (which is usually about 2 percent as the pole dries below fibre saturation), should be recognized if re-measuring diameter at a later date.

9.6

SYP

Diameter

The minimum diameter at 1.8m from the butt and at the top, for each length and class of pole, are listed in Appendix 3a. The diameter at 1.8m from the butt of a pole must be not more than 180mm or 20 percent larger than the specified minimum, whichever is greater. The top dimensional requirement shall apply at a point corresponding to the minimum length permitted for the pole.

Classification

The true diameter class must be determined as follows:

Measure the diameter at 1.8m from the butt. This dimension will determine the true class of the pole, provided that its top (measured at the minimum length point) meets or exceeds the minimum tip dimension for that pole class.

Otherwise, the diameter at the top will determine the true class, provided that the diameter at 1.8m from the butt does not exceed the specified minimum by more than 180mm or 20 percent, whichever is greater.

(see Appendix 3a for Classes)

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10 DECAY & INSECTS

10.1

BASE

Wood poles must be sound and free from decay and attack by insects.

Minor insect holes are acceptable providing these are either, not larger than 1.5mm in diameter and do not exceed 5 in number, or not larger than 1.0mm in diameter and do not exceed 20 in number (and are evenly distributed in any 100mm length or the wood pole).

There must not be more than 200 pinholes in total in any pole.

10.2

BT
&
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Poles must be of sound wood; rot or punks are not permitted.

In general, insect damage must be avoided, and poles with active insect infestation at or following final inspection are not acceptable.

Poles containing holes greater than 1.5mm diameter (shot holes or grub holes) are not acceptable.

Poles containing a number of pinholes (holes less than 1.5mm diameter) are acceptable within the following limitations.

- Pinholes are not permitted within 2m of the tip.
- The remaining surface area must not contain more than 50 pinholes evenly spaced in any 300mm section.
- The surface area of the pole must not contain more than 200 pinholes in total.

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Decay (other than Incipient Decay) is not permitted.

Incipient Decay is defined as: the early stage of decay that has not proceeded far enough to soften or otherwise perceptibly impair the hardness of the wood. Although the wood is invaded and some of its properties may have deteriorated, no visible change has occurred, with perhaps the exception of minor discoloration. Incipient decay can occur in living trees.

Insect damage, consisting of holes 2mm or less in diameter, or surface scoring or channelling is permitted. All other forms of insect damage are prohibited.

Explanation:

The bark beetle damage in this pole (below) is acceptable as long as there are no holes extending into the pole:



10.3

SYP

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11 KNOTS & KNOT CLUSTERS

All knots must be free from decay although dead knots are permitted.

The maximum dimension of knots, knotholes and knot clusters must be measured in the following manner (See also Figure 1):

- a) Individual knots or knot clusters - maximum diameter of knots or knot clusters expressed as a factor of the circumference of the wood pole at the point where the knot occurs;

and

- b) multiple knots, etc. - maximum sum of all the knot diameters in any 300mm length of the wood pole, expressed as a factor of the circumference of the wood pole at the mid-point of the 300mm length (e.g. factor = knot diameter in mm/circumference of pole at cross section in millimetres).

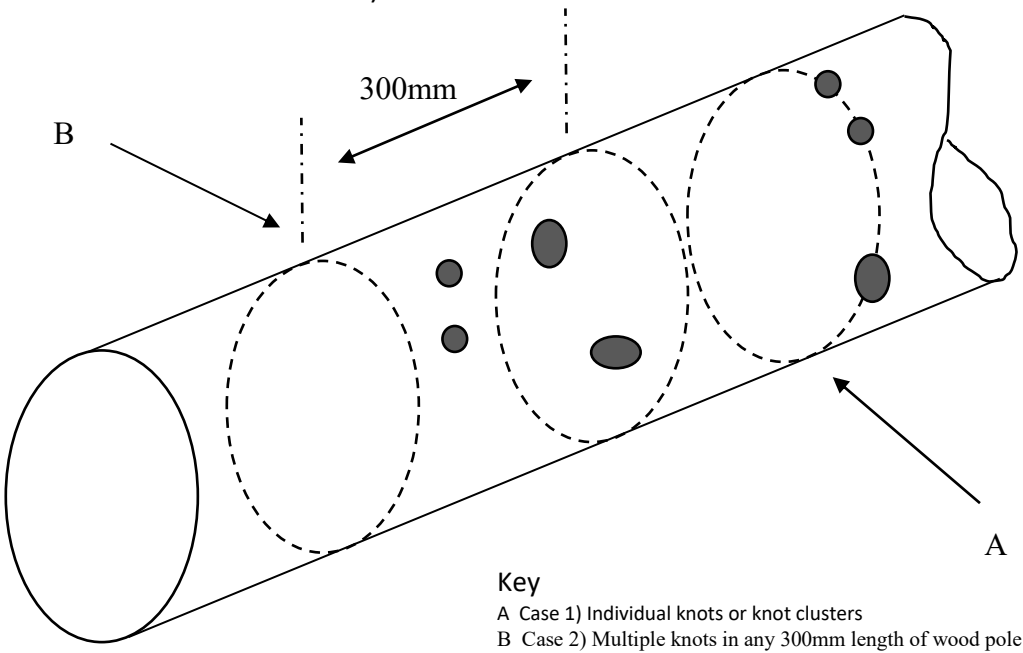


Figure 1 – Measurement of knots

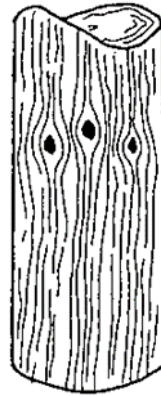
The dimension of a knot or knot cluster must be measured as the diameter of a knot measured on the surface of the pole and perpendicular to the axis of the pole.

Knot clusters must be treated as a single knot and must be differentiated in accordance with Figure 1 and Figure 2.

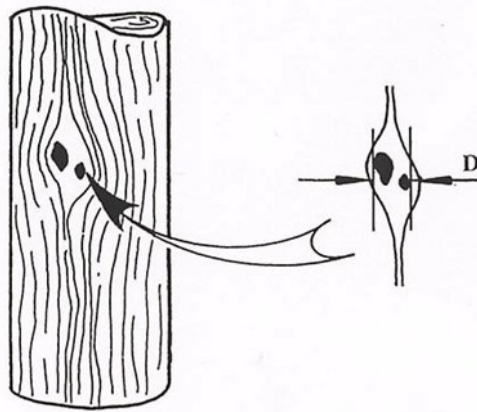
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BASE

Knots in timber



Treat knots separately



Treat knots as one

Figure 2 – Knots in timber

When two knots are in such close proximity that there is no grain recovery the knots must be treated as one.

An individual knot or knot cluster must have a dimension factor no greater than 0.064.

Multiple knots, in any 300 mm length of the wood pole, must have a dimension factor no greater than 0.24.

11.2

NGED

The minimum distance between adjacent rings of knots is 300mm. The distance must be measured between the edges of the knots.

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11.3

BT
&
CFR

Decayed or loose knots are not permitted.

Soft knots containing loose or soft fibres are permitted provided that the affected area does not exceed 15mm diameter and the depth is less than 10mm (depth to be measured with a BT Pole Probe or a handled 6.7mm diameter steel spike, the end 12mm of which must be shaped to a sharp point. Insertion force must be $12 \pm 1\text{kg}$)

The term knot in this clause includes knots and knot clusters.

At any cross section of the pole at 90° to the longitudinal axis of the pole, the sum of the surface diameters of any knots of which any visible part is present on the cross section must not exceed 0.2 times the circumference at that point.

The number of knots of which any visible part is present on the cross section must not exceed five.

For poles up to and including 13m length and for the bottom two-thirds of poles over 13m in length, no individual knot diameter measured at 90° to the longitudinal axis of the pole must exceed 0.065 times the pole circumference at that cross section or 50mm, whichever is the smaller.

For the top one-third of poles over 13m in length no knot diameter measured at 90° to the longitudinal axis of the pole must exceed 0.08 times the pole circumference at that cross section.

Depressions adjacent to knots in excess of 5mm in depth are not permitted.

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The diameter of any single knot and the sum of knot diameters in any 310mm section must not exceed the limits shown in the table below:

Maximum sizes permitted			
Diameter of any single knot (mm)			Sum of diameters of all knots (and cone holes, if applicable) greater than 13mm in any 310mm section (mm)
Length of Pole	Classes H6 to 3	Classes 4 to 10	All Classes
14.0m and shorter			
Lower half of length	80mm	50mm	1/3 of the average circumference of the same 310mm section or 200mm, whichever is greater, but not to exceed 310mm ¹⁾
Upper half of length	130mm	100mm	
15.0m and longer			
Lower half of length	100mm	100mm	1/3 of the average circumference of the same 310mm section or 250mm, whichever is greater, but not to exceed 360mm ¹⁾
Upper half of length	150mm	150mm	

11.4

SYP

¹⁾ Both upper and lower halves

In determining the sum of the knot diameters in any 310mm section, only those knots with diameters over 13mm whose pith centres fall within the section must be included in the sum, and the 310mm section must be located so as to include the maximum number of knots (i.e., the most severe condition).

Type II “decayed knots” are permitted. Type II knots have decayed fibres, but the decayed fibres do not extend into the pole’s centre.

Explanation

There are two types of decayed knots:

- Type I decayed knots have decayed fibres which extend deep into the pole and are associated with heart rot;
- Type II decayed knots are simply decayed knot fibres that do not extend deep into the pole and are not a cause for concern.

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Type I decayed knots are prohibited and are cause for rejection of a pole. Here is an example of a decayed knot that was determined to be associated with heart rot, causing this pole to be rejected.



This Type II knot does have decayed fibres, but they do not extend into the pole's centre, and is therefore permitted.



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12 FISSURES & SPLITS

12.1

BASE

Seasoning fissures along the grain of the wood are expected and are not recognised as defects providing they do not have a depth greater than half the diameter at any point along the wood pole when measured by inserting a 0.2mm feeler gauge as far as possible into the fissure and do not exceed 50% of the length of the wood pole

The combined width of all fissures at any section perpendicular to the longitudinal axis must not exceed 12mm.

12.2

BT
&
CFR

Seasoning shakes (or checks) along the grain are expected and are not recognised as defects within the following limits.

- Shakes must not have a depth greater than half the diameter at any point along the pole, when measured with a 0.2mm feeler gauge.
- No shake must be continuous from pole butt to tip.

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Splits in the Pole Top

A split or a combination of two single checks (each check terminating at the pith centre and separated by not less than 1/6 of the circumference) having one or both portions located in a vertical plane within 30 degrees of the top bolt hole shall not extend downward along the pole more than 150mm.

All other combinations of checks or a split shall not extend downward along the pole more than 310mm.

Splits in the Pole Butt

A split or a combination of two single checks, as defined above, shall not extend upward along the pole more than 610mm.

Explanation:

Splits are caused by mechanical stresses imposed upon a pole. However, in some cases, splits are indicated when two deep checks intersect near the centre of the pole.

Here is a photograph of a pole with a split top. The limitation of splits depends upon the location of the split in relation to the bolt hole, but in no case can extend down the pole more than 310mm.



13 CRACKS

13.1 **BASE** Cracks across poles or across the grain are not permitted.

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14 SLOPE OF GRAIN

14.1

BASE

The grain of the timber will normally be parallel to the longitudinal axis.

A slope of grain not exceeding 1:6 is acceptable.

There must be no significant changes in the slope of grain.

The slope of grain relative to the longitudinal axis must be measured over a minimum 1 m length of the wood pole.

EXAMPLE - A slope of 1 in 6 represents 1/6m (i.e. 167 mm) deviation over a 1.0m length along the axis of the pole.

The grain direction of the wood pole must be determined by one of the following methods from which the slope of grain must be calculated:

BASE

a) by taking a line parallel to the surface fissures;

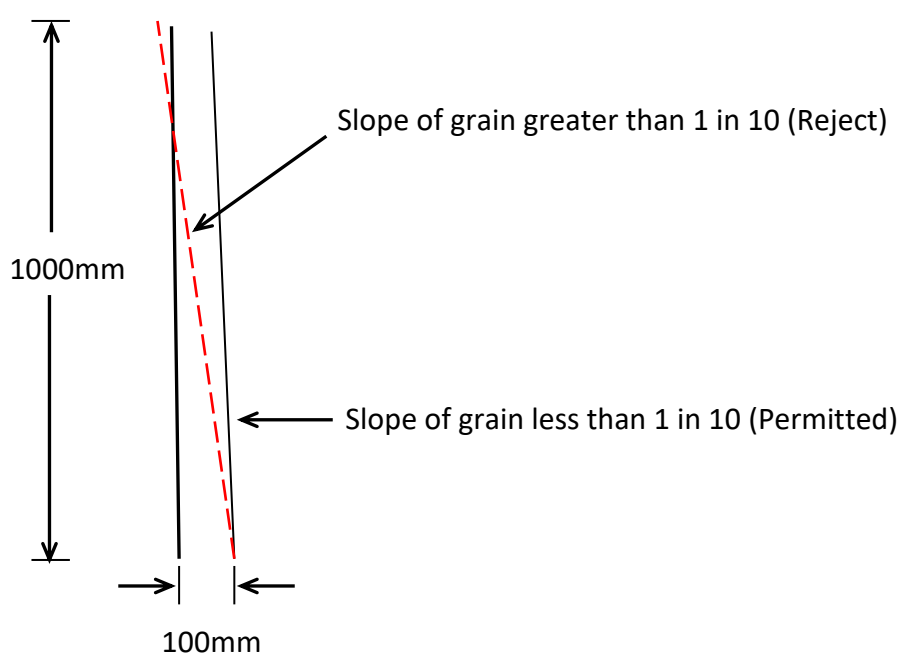
or

b) by the use of a grain detector (scribe).

14.2

BT
&
CFR

Spiral grain must not exceed 1 in 10 measured over any 1m along the pole. The slope is calculated by measuring 100mm perpendicular to the centre line of the pole, and 1000mm parallel to the centre axis (see below).



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Spiral grain (twist grain) is permitted as follows:

Length of pole	Maximum twist of grain permitted
9.1m and shorter	1 complete twist in any 3m
10.7m to 13.7m, inclusive	1 complete twist in any 5m
15.2m and longer	1 complete twist in any 6m

The following photograph is of a short section of a southern pine pole rejected because of spiral grain. Note the way in which the check is at a severe angle compared to the upper surface of the pole.

10.3 SYP



15 BLUE STAIN

15.1 **BASE** The presence of blue stain is not considered a defect.

16 MARKING

16.1 Refer to customer specific section of Module 5 (Pole Marking).

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17 POLE TESTS

17.1

BASE

The tests below must be carried out on any pole where there is a suspicion of decay, lamination or structural defect. The test must be repeated after final treatment, immediately prior to delivery. Tests must be carried out by a suitably qualified person, experienced enough to distinguish between acceptable (pass) and unacceptable (fail).

- The traditional hammer test for residual strength.
- A visual check for delamination.
- If necessary, supplemental core extraction and examination.

Any poles that fail either (or both) of the above tests must be rejected.

18 SAWING & MACHINING

18.1

BASE

The notching specified to accommodate brace blocks, key blocks or steelwork must be cut at $90^\circ \pm 5^\circ$ to the longitudinal axis and the flat face finished parallel to the longitudinal axis.

The crossarm scarfing must be at $90^\circ \pm 5^\circ$ to any sweep in the pole.

In the case of 'A' poles, the notching etc. must be cut as specified.

19 TWO MEMBER STRUCTURES

19.1

BASE

Poles to be used in the fabrication of 'H', 'A', 'Rutter', or 'Twin' structures must be straight and closely matched in their 'raw' dimensions, with fabrication bringing the structures to their final dimensions.

Where requested by customers, twin structures must be pre-assembled, complete with brace and key blocks, and if required, with steelwork and/or cross bonding.

Each component must be clearly and indelibly identified, as specified.

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Twin structures must be fabricated strictly to the designs, dimensions and tolerances specified in the controlled versions of the customer supplied drawings (and any accompanying notes).

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20 CONE HOLES (SYP ONLY)

Cone holes are typically less than one-half inch in diameter and those which are greater than one-half inch are measured along with knots and must not exceed the limits shown in the following table:

Length of Pole	Maximum sizes permitted		
	Diameter of any single cone hole (mm)		Sum of diameters of all cone holes, greater than 13mm in any 310mm section (mm)
	Classes H6 to 3	Classes 4 to 10	All Classes
14m and shorter			1/3 of the average circumference of the same 310mm section or 200mm, whichever is greater, but not to exceed 310mm ¹⁾
Lower half of length	80mm	50mm	
Upper half of length	130mm	100mm	
15m and longer			1/3 of the average circumference of the same 310mm section or 250mm, whichever is greater, but not to exceed 360mm ¹⁾
Lower half of length	100mm	100mm	
Upper half of length	150mm	150mm	

¹⁾ Both upper and lower halves

Explanation:

Cone holes are occasionally found in Southern Pines. These holes develop at the base of a pine cone that is retained on the main stem of the tree.

As the tree grows, the cone is pushed outward, leaving behind a hole filled with pitch which melts and drains during kiln drying.



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21 SCARS (SYP ONLY)

No pole is permitted to have a scar or turpentine cat face located within 610mm of the groundline.

Turpentine scars need be trimmed only to the extent necessary for examination for evidence of fungus infection and insect damage. Other sound scars are permitted elsewhere on the pole surface, provided they are smoothly trimmed and do not interfere with the cutting of any grain, and provided that:

1. The circumference at any point on trimmed surfaces located between the butt and 610mm below the groundline is not less than the minimum circumference specified at 1.8m from the butt for the class and length of the pole;
2. The depth of the trimmed scar is not more than 50mm, if the diameter is 250mm) or less, or 1/5 the pole diameter at the location of the scar if the diameter is more than 250mm.

Explanation:

21.1 SYP

'Scars' are essentially rindgalls and they are typically caused by the destruction of or trauma to the cambium of a living tree. The tree then grows new wood to cover the scar. This is a typical cat-face scar in southern pine:



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21 BUTT DEFECTS (SYP ONLY)

Hollowing in the butt caused by "splinter pulling" in felling the tree is permitted, provided that the area of such a hollow is less than 10% of the butt area.

Explanation:

Here is an example of splinter pulling in southern pine. It can be accepted, as the void area makes up less than 10% of the total area of the butt surface.



21.1 SYP

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22
SHELLING (SYP ONLY)

Shelling on the surface of the pole is limited to no more than 25mm in depth and must not exceed 1/3 of the pole’s circumference at the point of shelling.

Explanation:
The southern pine pole below shows the effects of shelling where the growth rings extend from an inner growth ring to the surface of the pole around a knot.



22.1
SYP

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APPENDIX 1 – ILLUSTRATIONS & DESCRIPTIONS OF POLE DEFECTS



Ambrosia Damage – Beetle damage characterised by round holes up to about 3mm in diameter; the tunnels are usually darkly stained (see Stain) not containing bore-dust (frass) and penetrating the timber mainly cross the grain. Ambrosia Beetles are commonly called PINHOLE BORERS or SHOTHOLE BORERS, according to the size of the holes. They are active in unseasoned timber only.



Bird Peck – A small hole or patch of distorted grain, sometimes associated with discolouration, attributed to the action of the birds.

IMAGE
NOT
AVAILABLE

Butt Sweeps – A deviation from straightness within 1.5m of the butt.



Compression Wood – Abnormal wood (reaction wood) formed typically on the lower sides of branches and of leaning or crooked trunks of softwood (coniferous) trees. Zones of compression wood are typically denser and darker than the surrounding tissue and have abnormally high longitudinal shrinkage tending to cause warping and splitting.



Growth Ring – A layer of wood produced in one growing season, seen as a ring in cross section



Grubhole – A worm-hole usually more than 3mm in diameter. Such holes are typically due to Beehole borers, Carpenter ants, Jewel beetles, Longhorn beetles or Woodwasps. They may be round or oval in section, with or without fibrous bore-dust (frass). This term does not normally apply to damage by organisms that attack timber in service.



Inbark – Bark in a flute or pocket or associated with a knot which has been partially or wholly enclosed by the growth of the tree; resin or gum may sometimes be present in the pocket.

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Knot – A portion of a branch enclosed in the wood by the natural growth of the tree.



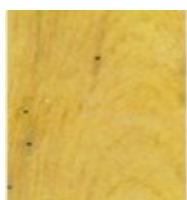
Knot Cluster – Set of knots close to each other and which can be considered as one defect. The grain of the pole is deviated around the set of knots and not around each separate knot.



Knot, Sound – A knot free from decay, solid across its face and at least as hard as the surrounding wood.



Knot, Unsound – A knot not solid across its face or softer than the surrounding wood due to decay or other defects.



Pin Hole – A worm-hole usually not more than 1.5mm in diameter. Such holes are typically due to Ambrosia beetles. Usually they are darkly stained (seen Stain) and do not contain bore-dust (frass). This term does not normally apply to damage by organisms that attack timber in service.



Punk – A wound or organic damage pocket in the wood where decay or Punk Knot infection has occurred and growth has failed to re-establish. A funnel-like cavity often filled or partially filled with non-fibrous matter. Normally caused by the fruiting body of *Trametes pini*, and is indicative of possible significant heart rot



Rate of Growth – The growth expressed as the number of growth rings per 25mm measured radially on the end grain of a log or piece of converted timber. Rate of growth must be measured at one end of the piece of timber must be expressed as the average number of growth rings intersected by a radial line 25mm long. Measurements must be made over a radial line as long as possible, up to 75mm.

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Rindgall – A surface wound that has been enclosed by the growth of the tree.



Rot (Decay) – Decomposition by fungi and other micro-organisms resulting in softening, progressive loss of strength and weight and often a change of texture and colour.



Sapstain (Blue Stain) – The most common form of sapstain producing a bluish discolouration.

Blue stain in the sapwood is not considered a defect. A discolouration of timber resulting from the growth of certain fungi that derive their nourishment from the cell contents but do not cause decomposition of the timber. It is principally confined to sapwood.



Sapwood – The outer layers of wood which, in the growing tree, contained living cells and reserve materials (e.g. starch); generally lighter in colour than heartwood though not always clearly differentiated.



Shake – A separation of the fibres along the grain due to stresses developing in the standing tree, or in felling or in seasoning.



13) Single Heart Shake



ii) Double Heart Shake

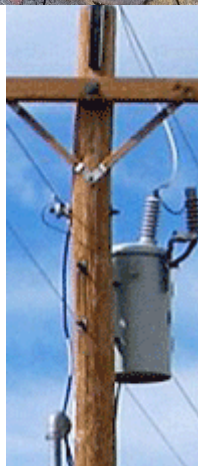
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Shake, Star – A number of heart shakes or less in the form of a star having 5 or more points.



Shake, Ring – A shake following a growth ring.



Short Crook – A localised deviation from straightness within any pole section of 1.5m or less.



Shothole – A worm-hole more than 1.5mm and not more than 3mm in diameter. Such holes may be due to Ambrosia beetles in which case they are usually darkly stained (See Stain) and not containing bore-dust (frass), or to Powder-post beetles, in which case they are unstained (see Stain) and may contain bore-dust (frass). This term does not usually apply to damage by organisms that attack timber in service.



Spiral Grain – Grain which follows a spiral course in one direction round the stem.

IMAGE
NOT
AVAILABLE

Turpentine Scars – Acid Face. An area in the lower portion of a Southern Pine pole where bark hack removal with acid applied has caused resin to flow. Cat Face. A depression in the surface of a Southern Pine pole resulting from a wood hack into the sapwood, where healing has not re-established the normal cross section of the pole.

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Worm – Any wood-boring organism (chiefly insects but including marine borers) that attacks wood producing holes or tunnels or similar damage.



Wood wasp Damage – Damage characterised by round holes up to 6mm in diameter; the tunnels are densely packed with bore-dust (frass). Found chiefly in softwood.



Worm Hole – A hole or tunnel, irrespective of size, caused by worm.

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APPENDIX 2 – RE-CLASSIFICATION CODES

BASE		
	Butt Sweep	BS
	Broken Pole	BP
	Crook	CROOK
	Compression (Stack)	CS
	Compression (Wood)	CW
	Damage	DAM
	De-Lamination	DEL
	Decay	ROT
	Dead Knot	DK
	Double Sweep	DS
	Damaged Tip	DT
	Fissure	F
	Fluted Butt	FB
	InBark	IB
	Insufficient Sapwood	IS
	Knot Cluster	KC
	Knot Depression	KD
	Oversized Knot	OSK
	Over Size Tip	OST
	Punk	P
	Oversized	POS
	Decay	R (ROT)
	Rindgall	RIND
	Rate of Growth	ROG
	Ring Of Knots	ROK
	Ring Shake	RS
	Split Butt	SB
	Saw Cut	SC
	Spiral Grain	SG
	Shake	SHAKE
	Shape	SHAPE
	Short Length	SHORT
	Split Tip	ST
	Sweep	SWEEP
	Under Size	US
	Under Size Tip	UST
	Worm	WO
	Wood Pecker	WP
	Wood Wasp	WW

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APPENDIX 3 – COMMONLY USED SIZES FOR WOOD POLES

(Sizes in **BOLD** are NPG ONLY)

BASE	Light Poles		Medium Poles		Medium Stout Poles		Stout Poles		Extra Stout Poles	
Length	Minimum Nominal Diameter at Tip	Minimum Nominal Diameter at 1.5m from Butt	Minimum Nominal Diameter at Tip	Minimum Nominal Diameter at 1.5m from Butt	Minimum Nominal Diameter at Tip	Minimum Nominal Diameter at 1.5m from Butt	Minimum Nominal Diameter at Tip	Minimum Nominal Diameter at 1.5m from Butt	Minimum Nominal Diameter at Tip	Minimum Nominal Diameter at 1.5m from Butt
m	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm
6	125	150								
7	125	160	140	200						
8	125	170	145	210						
8.5	125	180	150	215			190	265		
9	125	180	150	220			190	275	190 - 230	300 - 295
9.5	125	185	150	225			190	280	190	305
10	125	185	150	230	170	260	190	285	210 - 230	310 - 305
10.5	125	190	150	235	170	270	190	290	210	315
11	125	195	150	240	175	270	190	295	215 - 245	320
11.5	125	200	150	245	175	285	190	300	215	330
12	125	200	150	250	180	285	190	305	215 - 245	335
13	130	210	160	260	185	295	195	320	215 - 245	350
14			160	275			195	335	220 - 245	365
15			165	290			195	350	220 - 245	375
16			170	305			200	365	220 - 250	390
17			180	320			200	375	220 - 250	415
18			180	330			200	390	220 - 250	415
20			180	360			200	415	220 - 260	435
22			190	380			200	435	220 - 260	470
24							200	470		

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APPENDIX 3A – DIMENSIONS & CLASSIFICATION OF SOUTHERN PINE WOOD POLES

SYP	Class	H6	H5	H4	H3	H2	H1	1	2	3	4	5	6	7	9	10
Minimum diameter at top (mm)		315	299	283	267	251	236	220	204	185	169	153	137	121	121	95
Length of Pole (m)	Approximate Groundline Distance from Butt (m)*	Minimum diameter at 1.5m from butt (mm)														
7	1.4	362.3	347.7	331.8	315.9	299.9	283.4	265.4	249.5	233.8	217.9	200.6	180.6	171.5	153.3	119.7
8	1.6	383.2	367.3	351.4	333.7	317.8	298.5	280.1	262.5	246.7	229.9	214.0	192.3	182.1	162.8	
8.5	1.6	392.9	377.0	361.1	341.2	325.3	306.1	288.8	268.9	253.1	235.2	219.3	198.7	187.5	164.9	
9	1.7	402.7	386.7	370.8	348.7	332.8	313.6	297.4	275.3	259.6	240.5	224.6	205.2	192.8	166.9	
9.5	1.7	411.9	394.3	378.4	356.7	339.1	320.7	304.0	282.3	264.8	245.6	228.9	210.5	197.0		
10	1.7	420.9	401.3	385.4	364.7	345.1	327.7	310.0	289.3	269.8	250.6	232.8	215.4	201.0		
10.5	1.7	430.6	407.7	392.2	373.4	350.5	335.4	316.2	297.0	274.6	255.4	236.2	221.2			
11	1.7	438.1	416.2	399.7	379.8	358.0	340.7	321.5	302.3	279.9	260.7	241.5	224.3			
11.5	1.7	445.5	424.8	407.1	386.2	365.4	346.0	326.8	307.6	285.3	266.1	246.9	227.4			
12	1.8	453.0	433.3	414.6	392.6	372.9	351.3	332.1	312.9	290.6	271.4	252.2	230.6			
12.5	1.8	460.4	440.6	421.4	399.6	379.7	357.3	337.4	318.3	295.2	276.0	256.8	234.4			
13	1.9	467.9	446.9	427.8	407.0	386.1	363.7	342.7	323.6	299.4	280.3	261.1	238.6			
13.5	2.0	475.8	452.9	434.6	415.4	392.9	370.5	348.0	329.3	303.7	284.5	265.7				
14	2.0	482.1	460.3	439.8	420.7	398.2	375.8	353.4	333.5	307.9	288.7	268.9				
15	2.1	494.8	475.2	450.4	431.2	408.8	386.4	364.0	342.0	316.4	297.2	275.2				
16	2.2	506.9	486.1	462.1	441.3	417.2	396.4	372.4	348.4	324.4	305.2					
17	2.3	518.5	496.1	473.3	451.3	425.7	406.1	380.4	354.8	332.4	312.4					
18	2.4	529.1	506.7	481.1	462.6	436.9	415.2	388.9	363.2	341.5	315.2					
19	2.5	541.4	517.3	491.6	469.3	445.4	419.7	397.3	371.7	346.1	327.2					
20	2.5	547.9	531.4	505.7	486.0	451.9	428.9	409.1	386.1	357.9						
21	2.6	560.6	538.2	512.6	487.0	472.8	435.8	410.2	381.3	367.2						
22	2.9	563.9	541.5	521.1	495.4	466.6	444.2	423.8	395.0	366.2						

* The figures in this column are not recommended embedment depths; rather, these values are intended for use only when a definition of groundline is necessary in order to apply requirements relating to scars, straightness, etc.

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APPENDIX 4 – AVERAGE CALLIPER CUBE AND WEIGHTS (PINUS SYLVESTRIS)

BASE		Average Caliper Cube (see note 2) and Weight (see note 4)								
Length of Pole (m)	Light		Medium		Medium Stout		Stout		Extra Stout	
	Cubic (m)	Weight (kg)	Cubic (m)	Weight (kg)	Cubic (m)	Weight (kg)	Cubic (m)	Weight (kg)	Cubic (m)	Weight (kg)
8.0	0.199	143								
8.5	0.215	157	0.292	213			0.417	305		
9.0	0.225	163	0.315	229			0.463	335		
9.5	0.249	183	0.346	249			0.514	371		
10.0	0.270	198	0.360	259	0.448	336	0.550	395	0.555	416
10.5	0.293	213	0.394	284	0.449	374	0.595	427		
11.0	0.320	234	0.430	310	0.521	391	0.653	472	0.657	493
11.5	0.361	264	0.462	335	0.595	446	0.688	498		
12.0	0.392	284	0.490	356	0.620	465	0.743	539	0.757	568
13.0	0.437	315	0.576	417	0.710	532	0.828	599	0.866	649
14.0			0.643	467			0.949	686	0.997	748
15.0			0.765	554			1.057	762	1.103	828
16.0			0.869	630			1.174	848	1.237	928
17.0			0.995	721			1.317	950	1.426	1070
18.0			1.148	828			1.536	1110	1.505	1129
20.0			1.379	996			1.852	1340	1.775	1331
22.0			1.540	1110			2.149	1550	2.166	1624
24.0							2.254	1630		

NOTES:

13. The terms light, medium, and stout, refer to Annex A.

2. The calculation of the cubic contents of each pole has been based on the mean diameters taken by caliper.

3. The cubic content of non-standard poles must be calculated using the mean diameter midway between the butt and the tip.

4. The average density of creosoted poles is assumed to be 750 kg/m³ for calculation purposes.

5. For other species differing densities and / or tapers may apply.

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APPENDIX 5 – PERMITTED KNOT SIZES FOR REC

REC			
Pole Diameter (mm)	Circumference	Individual knots & knot clusters Max dia (mm)	Multiple knots in any 300mm length of pole & ring of knots (for all pole lengths) Max sum of knot dia (mm)
125	392.70	25.13	94.2
130	408.41	26.14	98.0
135	424.11	27.14	101.8
140	439.82	28.15	105.6
145	455.53	29.15	109.3
150	471.24	30.16	113.1
155	486.95	31.16	116.9
160	502.65	32.17	120.6
165	518.36	33.18	124.4
170	534.07	34.18	128.2
175	549.78	35.19	131.9
180	565.49	36.19	135.7
185	581.19	37.20	139.5
190	596.90	38.20	143.3
195	612.61	39.21	147.0
200	628.32	40.21	150.8
205	644.03	41.22	154.6
210	659.73	42.22	158.3
215	675.44	43.23	162.1
220	691.15	44.23	165.9
225	706.86	45.24	169.6
230	722.57	46.24	173.4
235	738.27	47.25	177.2
240	753.98	48.25	181.0
245	769.69	49.26	184.7
250	785.40	50.27	188.5
255	801.11	51.27	192.3
260	816.81	52.28	196.0
265	832.52	53.28	199.8
270	848.23	54.29	203.6
275	863.94	55.29	207.3
280	879.65	56.30	211.1
285	895.35	57.30	214.9
290	911.06	58.31	218.7
295	926.77	59.31	222.4
300	942.48	60.32	226.2
305	958.18	61.32	230.0

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REC		Individual knots & knot clusters Max dia (mm)	Multiple knots in any 300mm length of pole & ring of knots (for all pole lengths) Max sum of knot dia (mm)
Pole Diameter (mm)	Circumference		
310	973.89	62.33	233.7
315	989.60	63.33	237.5
320	1005.31	64.34	241.3
325	1021.02	65.35	245.0
330	1036.72	66.35	248.8
335	1052.43	67.36	252.6
340	1068.14	68.36	256.4
345	1083.85	69.37	260.1
350	1099.56	70.37	263.9
355	1115.26	71.38	267.7
360	1130.97	72.38	271.4
365	1146.68	73.39	275.2
370	1162.39	74.39	279.0
375	1178.10	75.40	282.7
380	1193.80	76.40	286.5
385	1209.51	77.41	290.3
390	1225.22	78.41	294.1
395	1240.93	79.42	297.8
400	1256.64	80.42	301.6
405	1272.34	81.43	305.4
410	1288.05	82.44	309.1
415	1303.76	83.44	312.9
420	1319.47	84.45	316.7
425	1335.18	85.45	320.4
430	1350.88	86.46	324.2
435	1366.59	87.46	328.0
440	1382.30	88.47	331.8
445	1398.01	89.47	335.5
450	1413.72	90.48	339.3
455	1429.42	91.48	343.1
460	1445.13	92.49	346.8
465	1460.84	93.49	350.6
470	1476.55	94.50	354.4
475	1492.26	95.50	358.1
480	1507.96	96.51	361.9
485	1523.67	97.51	365.7
490	1539.38	98.52	369.5
495	1555.09	99.53	373.2
500	1570.80	100.53	377.0

(Sum of knots – for intermediate sizes, calculate maximum sum as Circumference *24 /100)

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APPENDIX 6 – PERMITTED KNOT SIZES FOR BT & CITYFIBRE (CFR)

BT/CFR				
Pole Dia (mm)	Circumference	Max knot dia (mm) for poles up to 13.0m long and bottom 2/3 rd s of poles over 13.0m	Max knot dia (mm) for top 1/3 rd of poles over 13.0m long	Max sum of knot dia (mm) for ring of knots for all pole lengths 5 knots max in ring
125	392.70	25.53	31.42	78.54
130	408.41	26.55	32.67	81.68
135	424.11	27.57	33.93	84.82
140	439.82	28.59	35.19	87.96
145	455.53	29.61	36.44	91.11
150	471.24	30.63	37.70	94.25
155	486.95	31.65	38.96	97.39
160	502.65	32.67	40.21	100.53
165	518.36	33.69	41.47	103.67
170	534.07	34.71	42.73	106.81
175	549.78	35.74	43.98	109.96
180	565.49	36.76	45.24	113.10
185	581.19	37.78	46.50	116.24
190	596.90	38.80	47.75	119.38
195	612.61	39.82	49.01	122.52
200	628.32	40.84	50.27	125.66
205	644.03	41.86	51.52	128.81
210	659.73	42.88	52.78	131.95
215	675.44	43.90	54.04	135.09
220	691.15	44.92	55.29	138.23
225	706.86	45.95	56.55	141.37
230	722.57	46.97	57.81	144.51
235	738.27	47.99	59.06	147.65
240	753.98	49.01	60.32	150.80
245	769.69	50.00	61.58	153.94
250	785.40	50.00	62.83	157.08
255	801.11	50.00	64.09	160.22
260	816.81	50.00	65.34	163.36
265	832.52	50.00	66.60	166.50
270	848.23	50.00	67.86	169.65
275	863.94	50.00	69.12	172.79
280	879.65	50.00	70.37	175.93
285	895.35	50.00	71.63	179.07
290	911.06	50.00	72.88	182.21
295	926.77	50.00	74.14	185.35
300	942.48	50.00	75.40	188.50

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BT/CFR		Max knot dia (mm) for poles up to 13.0m long and bottom 2/3 rd s of poles over 13.0m	Max knot dia (mm) for top 1/3 rd of poles over 13.0m long	Max sum of knot dia (mm) for ring of knots for all pole lengths 5 knots max in ring
Pole Dia (mm)	Circumference			
305	958.18	50.00	76.65	191.64
310	973.89	50.00	77.91	194.78
315	989.60	50.00	79.17	197.92
320	1005.31	50.00	80.42	201.06
325	1021.02	50.00	81.68	204.20
330	1036.72	50.00	82.94	207.34
335	1052.43	50.00	84.19	210.49
340	1068.14	50.00	85.45	213.63
345	1083.85	50.00	86.71	216.77
350	1099.56	50.00	87.96	219.91
355	1115.26	50.00	89.22	223.05
360	1130.97	50.00	90.48	226.19
365	1146.68	50.00	91.73	229.34
370	1162.39	50.00	92.99	232.48
375	1178.10	50.00	94.25	235.62
380	1193.80	50.00	95.50	238.76
385	1209.51	50.00	96.76	241.90
390	1225.22	50.00	98.02	245.04
395	1240.93	50.00	99.27	248.19
400	1256.64	50.00	100.53	250.00
405	1272.34	50.00	101.79	250.00
410	1288.05	50.00	103.04	250.00
415	1303.76	50.00	104.30	250.00
420	1319.47	50.00	105.56	250.00
425	1335.18	50.00	106.81	250.00
430	1350.88	50.00	108.07	250.00
435	1366.59	50.00	109.33	250.00
440	1382.30	50.00	110.58	250.00
445	1398.01	50.00	111.84	250.00
450	1413.72	50.00	113.10	250.00
455	1429.42	50.00	114.35	250.00
460	1445.13	50.00	115.61	250.00
465	1460.84	50.00	116.87	250.00
470	1476.55	50.00	118.12	250.00
475	1492.26	50.00	119.38	250.00
480	1507.96	50.00	120.64	250.00
485	1523.67	50.00	121.89	250.00
490	1539.38	50.00	123.15	250.00
495	1555.09	50.00	124.41	250.00
500	1570.80	50.00	125.66	250.00

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Documents Replaced by this Training Guide

Document Ref.	Document Title
QTG35	BT Maximum Knot Size
QTG49	Permitted Knot Sizes 4388
QTG20	Pole Dressing & Fabrication

Updates		
Date	Amendment	By
4-12-2018	Amended during tender preparation.	DL
16-10-2019	Amended during system rationalisation exercise.	DL
13-5-2020	Amended to incorporate NBI requirements.	DL
27-11-2020	Amended to incorporate CityFibre requirements.	DL
12-7-2021	Amended reference from QTG55 to Module 5 in Sect. 16.	DL
6-9-2021	Added new WPD requirement for ROG measurement.	DL
5-10-2021	Updated with requirements from tenders.	DL
23-2-2022	BT knot table amended to revise permitted knot sizes in top 3 rd of poles >13.0m	DL
30-3-2022	Requirement to include sapwood in knot dimensions for BT removed.	DL
18-8-2022	All references to WPD changed to NGED to comply with EE SPEC: 129/2.	DL
10-11-2022	Maximum permissible tip size for BT stout poles increased from 230 to 250mm.	DL
1-6-2023	Nominal maximum dimensions added for NGED poles.	DL
23-10-2025	Full requirements for Southern Yellow Pine poles added	DL